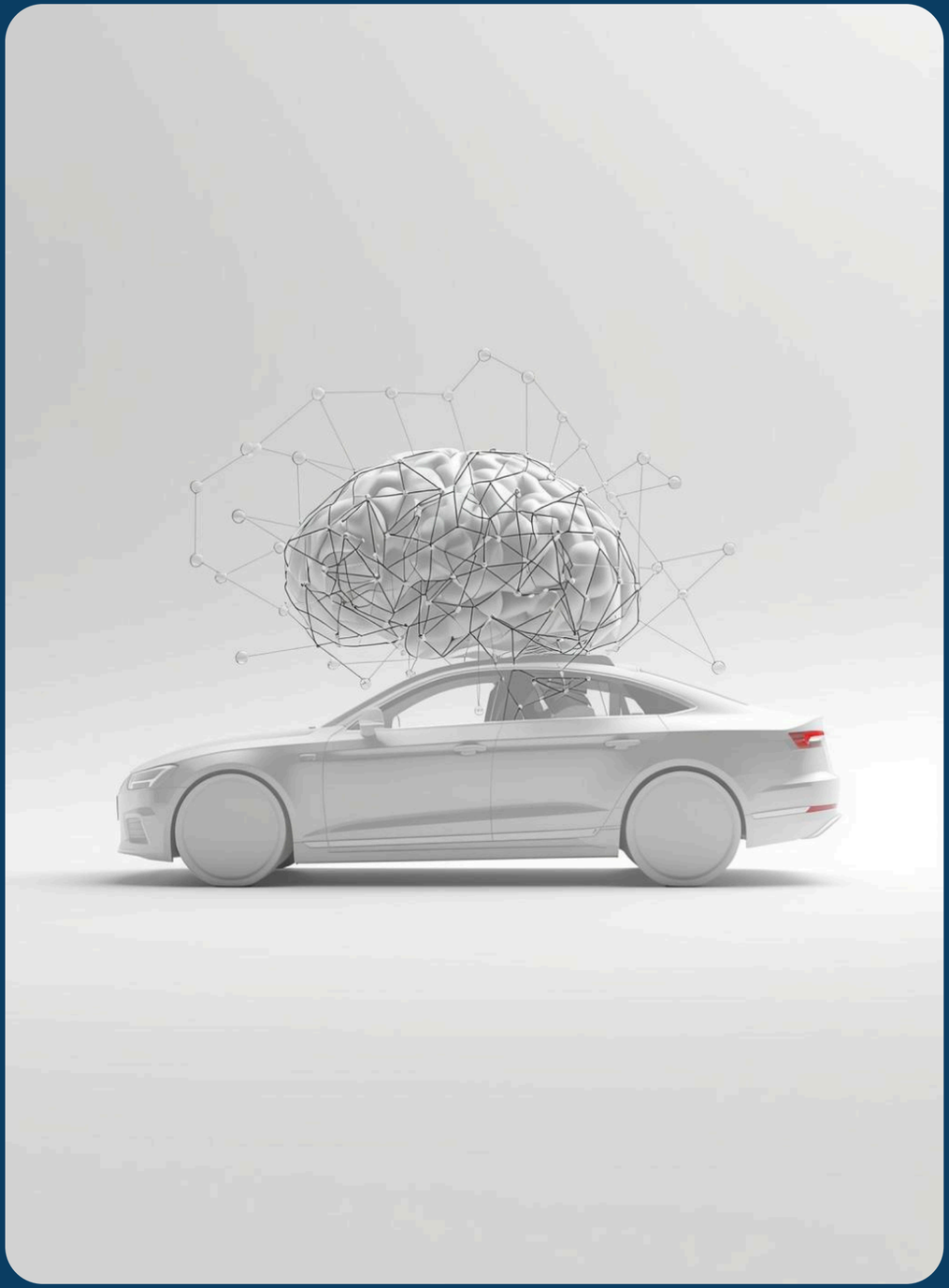
AI-Powered Driver Drowsiness Detection and Rescue System



Project Abstract

The rapid expansion of global road transportation and increasing vehicle density have intensified the need for autonomous safety systems to detect accidents and reduce response time. This project presents a real-time Automatic Vehicle Accident Detection and Rescue System designed to identify sudden impacts or abnormal vehicle movements using sensors. The system immediately activates a buzzer, displays alerts on an LCD, and sends notifications to enable quick emergency response and enhance road safety.

Team Overview

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Introduction

Road accidents caused by driver fatigue are a **critical issue** that necessitates real-time monitoring and detection systems.

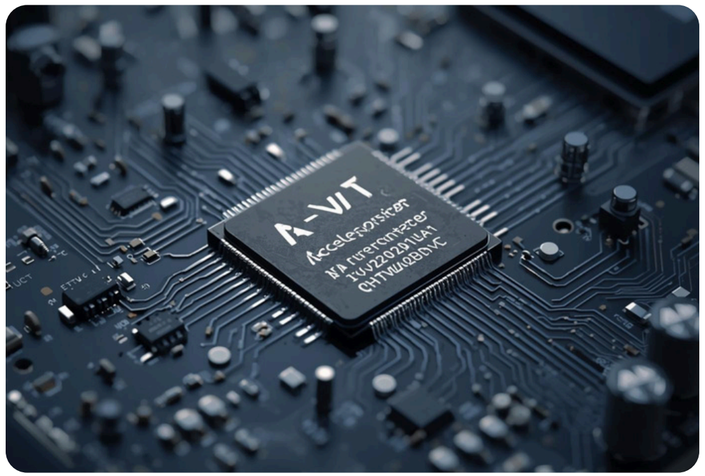
Existing System Limitations

GPS Tracking

Relies on location data but lacks drowsiness monitoring capability.

Accelerometer Use

Measures motion but does not provide driver fatigue detection.

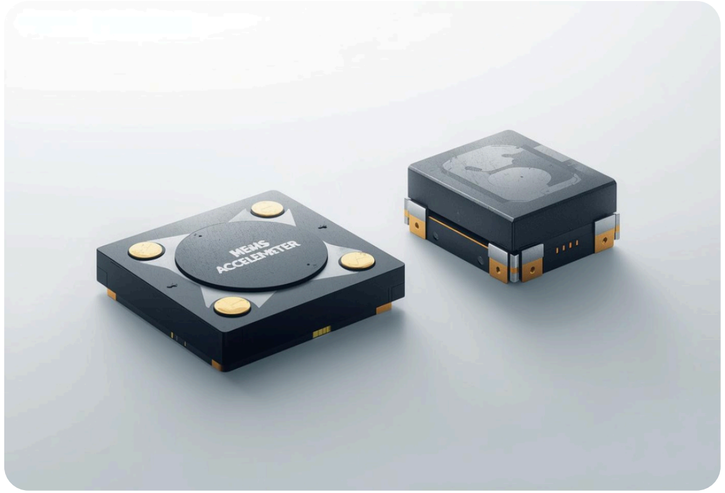


Proposed System

Real-Time Eye Detection for Alertness

AI-Powered Driver Monitoring System

Advanced Sensor Technologies for Safety



AI-Powered Driver Monitoring System

Explore the Future

Advanced Sensor Technologies for Safety

Discover More

Real-Time Eye Detection for Alertness

Stay Alert

Project Overview

Video Acquisition

Captures real-time footage to monitor driver behavior and alert drowsiness.

Drowsiness detection & Alert generation

Continuously records the driver's face to analyze eye activity and head movements.

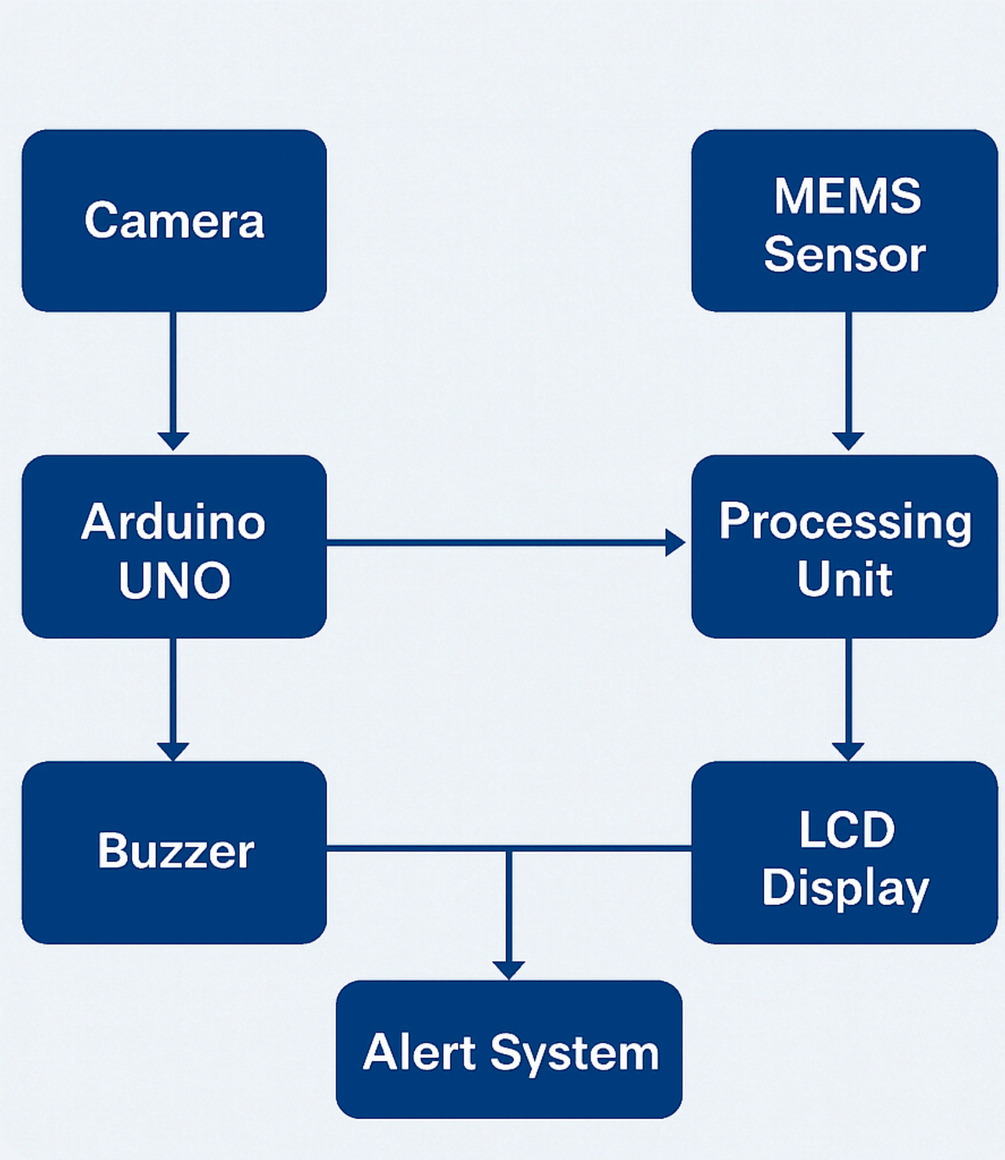
Image Preprocessing

Applies techniques to enhance video quality and prepare for analysis.

System Overview

Architecture Diagram

The system architecture shows how data flows between the components to detect accidents and trigger alerts. The camera captures real-time information and sends it to the Arduino Uno, while the MEMS sensor provides motion and impact data to the processing unit. The processing unit analyzes this data and displays the results on an LCD screen. If an accident is detected, the buzzer and alert system are activated to notify people and initiate emergency response.



Implementation of the Smart System

Continuous Monitoring

The system continuously monitors the driver's behavior using an onboard camera to detect signs of drowsiness.

Sensor Integration

Data is collected from MEMS accelerometer and ultrasonic sensors to assess vehicle status and surroundings in real-time.

AI Processing

AI algorithms analyze input data to identify drowsiness indicators and potential accident scenarios for immediate alerts.

Alert Generation

Alerts are triggered instantly upon detecting drowsiness or an accident, ensuring rapid response from emergency services.

Essential Hardware Components for System

Arduino UNO

A versatile microcontroller used for processing sensor data and executing program logic in real-time.

MEMS Accelerometer

Measures vehicle acceleration and detects sudden movements, crucial for identifying potential accidents.

Ultrasonic Sensor

Detects obstacles in the vehicle's path, enhancing safety by preventing collisions through alert mechanisms.

Camera Module

Captures real-time video for driver monitoring, enabling drowsiness detection through advanced image processing techniques.

LCD Display

Displays real-time system status and accident information.

Buzzer

Generates an immediate audible alert during accidents or emergencies, ensuring quick attention and response.

Software Requirements for the Project

Arduino IDE

The primary tool for programming the Arduino microcontroller, facilitating efficient development and debugging of the code.

Embedded C

The programming language used for the implementation of algorithms that control hardware functionality and processing.

Computer Vision & AI Libraries

Algorithms enabling real-time face and eye detection to assess driver drowsiness and enhance monitoring accuracy.

IoT Bot Application

IoT-enabled real-time monitoring and instant alert system for enhanced vehicle safety and remote emergency response.

Conclusion

This project aims to enhance **road safety** through effective driver monitoring and accident detection.

Thank you!

For questions, reach out at: m.chaitanyareddy999@gmail.com